

SURE ProEd

Skill Up-Gradation for Rural Youth Empowerment

Antarctica Data Observatory

Power BI Dashboard Project Report

An Advanced Analytical Dashboard for Real-Time Antarctic Cryosphere Intelligence

Student Name	Thota Sushma
Domain	Power BI
Project Title	Antarctica Data Observatory
Tool Used	Microsoft Power BI Desktop
Data Sources	SCAR, NASA GRACE/GRACE-FO, NSIDC, IMBIE, NOAA
Visualization Types	KPI Cards, Line Charts, Bar Charts, Maps, Slicers, Gauges
Organization	SURE ProEd – Skill Up-Gradation for Rural Youth Empowerment
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Confidential – Academic Project Report | Final Project Submission

1. Abstract

The Antarctica Data Observatory is an advanced Power BI dashboard project designed to visualize, monitor, and analyze scientific data from the world's coldest and most climate-critical continent. Antarctica plays a pivotal role in regulating global sea levels, ocean circulation, and planetary temperatures. Yet its vast, remote nature makes real-time data interpretation challenging without powerful visualization tools.

This project consolidates multi-source, peer-reviewed datasets from globally recognized scientific institutions — including the Scientific Committee on Antarctic Research (SCAR), NASA's GRACE and GRACE-FO satellite missions, the National Snow and Ice Data Center (NSIDC), the Ice Sheet Mass Balance Inter-comparison Exercise (IMBIE), and NOAA Climate.gov — into a single, interactive Power BI dashboard.

The dashboard delivers real-time-style cryosphere intelligence through an extensive array of visualizations: KPI summary cards, multi-series line charts tracking ice mass loss from 1992 to 2024, bar charts showing annual sea ice extent records, gauge visuals for sea level contribution, choropleth maps of Antarctic regions, and interactive slicers for year-range filtering. Advanced DAX measures are employed to compute cumulative ice loss, year-over-year change rates, moving averages, and anomaly detection flags.

Key findings surfaced by the dashboard include the tripling of Antarctic ice loss rates from approximately 60 Gt/yr in the early 1990s to over 220 Gt/yr by 2024, four consecutive record-low sea ice minimum extents (2022–2025), and the alarming instability of West Antarctic glaciers such as Thwaites and Pine Island. The project demonstrates how Power BI can transform complex scientific datasets into accessible, high-impact analytical stories.

2. Introduction

Antarctica, the southernmost continent, is Earth's largest reservoir of fresh water, containing approximately 26.5 million km³ of ice. It holds 90% of the world's ice mass and over 70% of its fresh water. Changes in the Antarctic ice sheet directly influence global mean sea levels, ocean thermohaline circulation, and regional climate patterns worldwide. Understanding these changes is not merely an academic exercise — it has profound implications for coastal communities, ecosystems, and global policy.

In recent decades, satellite-based monitoring programs have generated vast quantities of cryosphere data. However, this data is often fragmented across multiple scientific databases, published in technical papers, or presented in raw numerical formats that are inaccessible to non-specialist audiences. There exists a significant gap between data availability and data comprehension.

This project addresses that gap by leveraging Microsoft Power BI — a leading business intelligence and data visualization platform — to create the Antarctica Data Observatory: a comprehensive, multi-page interactive dashboard that translates complex scientific datasets into clear, compelling visual narratives.

The primary objectives of this project are:

- To aggregate and clean multi-source Antarctic scientific datasets into structured Power BI data models.
- To design visually rich, informative dashboards covering ice mass balance, sea ice extent, temperature trends, sea level contribution, and biodiversity records.
- To implement advanced DAX (Data Analysis Expressions) calculations for dynamic KPIs, trend analysis, and anomaly flagging.
- To demonstrate the capability of Power BI as a scientific data visualization tool beyond traditional business use cases.
- To present data in a manner that is accessible to both scientific and general audiences.

Data for this project is sourced exclusively from peer-reviewed scientific publications and official monitoring agencies, ensuring accuracy, credibility, and reproducibility. The SCAR Antarctic Master Directory alone contains over 7,700 dataset descriptions from 25 countries, reflecting the depth of scientific data available for this topic.

3. Scope of the Project

The Antarctica Data Observatory encompasses a broad and deeply analytical scope, covering five major thematic domains of Antarctic scientific monitoring. Each domain is represented as a dedicated page within the Power BI report, with cross-page filtering enabled through shared slicers.

3.1 Ice Sheet Mass Balance

Tracking cumulative ice loss from the three major Antarctic regions — West Antarctica, East Antarctica, and the Antarctic Peninsula — from 1992 to 2024. Data sourced from IMBIE (Ice Sheet Mass Balance Inter-comparison Exercise), incorporating 50 independent satellite measurement estimates.

3.2 Sea Ice Extent Monitoring

Annual minimum and maximum sea ice extent records from 1979 to 2025, sourced from NSIDC's Sea Ice Index. Includes detection of record-breaking years, deviation from 1981–2010 median baseline, and trend analysis.

3.3 Temperature & Climate Trends

Surface air temperature records from major Antarctic research stations (Vostok, McMurdo, Faraday-Vernadsky, Byrd, South Pole) using SCAR's Met-READER database. Includes extreme temperature event mapping and regional warming trend analysis.

3.4 Sea Level Contribution

Antarctica's contribution to global mean sea level rise by time period, computed from IMBIE mass balance data. Visualizes the acceleration from 1.1 mm per period in 1992–97 to over 2.6 mm per period in 2020–24.

3.5 SCAR Scientific Databases Overview

An overview dashboard of SCAR's major data repositories including the AMD (7,700+ dataset descriptions), biodiversity.aq (8,200+ marine species), the International Iceberg Database (35,000+ records), and the Continuous Plankton Recorder Database.

3.6 Future Enhancement Scope

- Integration with live SCAR API feeds for real-time data refresh in Power BI Service.
- Machine learning-powered anomaly detection using Azure ML integration.
- Predictive ice loss modeling using Power BI's AI visuals (Key Influencers, Decomposition Tree).
- Mobile-optimized report layout for Power BI Mobile app.
- Publication as a public Power BI embedded report for open scientific access.

4. Existing System vs. Proposed System

4.1 Existing System – Limitations

- Antarctic scientific data is scattered across dozens of separate databases (SCAR AMD, NSIDC, NASA, NOAA) with no unified access point.
- Raw data is published in CSV, NetCDF, and technical PDF formats inaccessible to non-specialist audiences.
- Existing visualizations are static charts in journal publications — not interactive or filterable.
- No integrated cross-domain view linking ice mass loss, sea ice, temperature, and sea level rise in a single dashboard.
- No dynamic KPI tracking — scientists must manually calculate year-over-year change and cumulative metrics.
- Limited accessibility — data interpretation requires specialist knowledge of climate science and statistics.

4.2 Proposed System – Key Features Implemented

Feature	Implementation	Benefit
Multi-Source Data Integration	Power Query ETL from SCAR, NASA, NSIDC, IMBIE	Unified data model from 5+ scientific agencies
Ice Mass Balance Tracking	Multi-series line chart with DAX cumulative measures	Clear visualization of 30+ year ice loss acceleration
Sea Ice Extent Dashboard	Bar + line combo chart with anomaly highlights	Instant identification of record-breaking years
KPI Summary Cards	DAX measures: total loss, current rate, sea level rise	At-a-glance scientific intelligence
Regional Analysis	Filled map + clustered bar chart by Antarctic region	West vs East Antarctica vs Peninsula comparison
Temperature Trends	Multi-station line chart with trend lines	Station-by-station warming rate visualization
Advanced DAX Calculations	YoY%, moving averages, anomaly flags, cumulative sums	Dynamic analytics beyond static charts
Interactive Slicers	Year range, region, station, and dataset filters	User-controlled exploration of 45+ years of data
Gauge Visuals	Sea level contribution gauge with target thresholds	Intuitive progress-toward-threshold visualization
Drill-through Pages	Glacier detail pages (Thwaites, Pine Island)	Deep-dive capability from summary to detail

4.3 Comparison Summary

Aspect	Existing Approach	Proposed System
Data Access	Fragmented across 5+ separate databases	Unified Power BI data model
Interactivity	Static charts in publications	Fully interactive with slicers & drill-through
Audience	Climate scientists only	Scientists and general public
Cross-Domain View	Not available	5 integrated thematic dashboards
Real-Time KPIs	Manual calculation required	Automated DAX measures
Trend Analysis	Manual statistical methods	Built-in trend lines + moving averages
Accessibility	Requires specialist knowledge	Intuitive visual storytelling
Update Capability	Static — new publications required	Refreshable data model

5. Data Sources & Datasets

All datasets used in this project are sourced from internationally recognized scientific institutions and peer-reviewed publications. Each dataset is publicly available and has been verified for accuracy and scientific credibility.

5.1 Primary Data Sources

Source / Agency	Dataset Used	Coverage	Access
IMBIE (Ice Sheet Mass Balance)	Antarctic ice mass change 1992–2020	28 years, 3 regions	essd.copernicus.org
NASA GRACE / GRACE-FO	Monthly ice mass anomaly 2002–2025	23 years, continental	svs.gsfc.nasa.gov
NSIDC Sea Ice Index	Annual min/max sea ice extent 1979–2025	46 years	nsidc.org
SCAR Met-READER	Monthly station temperatures, multiple stations	1957–present	scar.org/reader
SCAR Antarctic Master Directory	7,700+ dataset descriptions, 25 countries	Multi-decade	search.earthdata.nasa.gov
NOAA Climate.gov	Sea ice event analysis, extreme records	2002–2025	climate.gov
SCAR Biodiversity.aq	Marine & terrestrial species records	Ongoing	biodiversity.aq
BAS (British Antarctic Survey)	Climate change reports, temperature trends	1950–present	bas.ac.uk

5.2 Key Scientific Values Used in Dashboard

Metric	Value	Source	Year
Total Antarctic ice lost (1992–2017)	2,720 ± 1,390 Gt	IMBIE / Nature 2018	2018
Annual ice loss rate (2002–2025)	~135 Gt/yr	NASA GRACE-FO	2025
Sea ice minimum – record low	1.79 M km ² (21 Feb 2023)	NSIDC	2023
Sea ice minimum – 2025	1.98 M km ² (1 Mar 2025)	NSIDC	2025
West Antarctica loss rate	–82 ± 9 Gt/yr	IMBIE 2023	2023
Peninsula loss rate	–13 ± 5 Gt/yr	IMBIE 2023	2023

East Antarctica mass change	+3 ± 15 Gt/yr	IMBIE 2023	2023
Sea level contribution (1992–2017)	7.6 ± 3.9 mm	IMBIE / Nature 2018	2018
Faraday-Vernadsky warming rate	+0.46°C/decade	Turner et al. 2020	2020
Byrd Station warming (since 1958)	+2.4°C total	NCAR / Ohio State	2013
Record low temp (Vostok, 1983)	−89.2°C	WMO Verified	1983
Record high temp (Signy, 1982)	+19.8°C	WMO Verified	1982

6. Dashboard Architecture & Design

The Antarctica Data Observatory is structured as a multi-page Power BI report with a consistent navigation model, shared color theme, and cross-page filtering. The architecture follows a layered approach: data ingestion, transformation, modeling, and presentation.

6.1 Power BI Report Structure

Page No.	Page Name	Primary Visuals	Data Source
1	Executive Overview	KPI Cards, Ice Loss Summary, Sea Ice Donut	IMBIE, NSIDC, NASA
2	Ice Sheet Mass Balance	Multi-series Line Chart, Regional Bar Chart, Gauge	IMBIE 1992–2024
3	Sea Ice Extent	Annual Bar Chart, Min/Max Line, Anomaly Highlight	NSIDC 1979–2025
4	Temperature Trends	Multi-Station Line, Heat Map, Station Cards	SCAR Met-READER
5	Sea Level Impact	Stacked Area, Acceleration Bar, Contribution Gauge	IMBIE, NASA
6	Regional Analysis	Antarctica Filled Map, Glacier Drill-through	IMBIE, NASA
7	SCAR Databases	KPI Cards, Database Cards, Treemap	SCAR AMD, biodiversity.aq
8	Critical Events Timeline	Milestone Visual, Event Cards, Year Filter	Multiple SCAR sources

6.2 Data Model Design

The Power BI data model follows a star schema with the following tables:

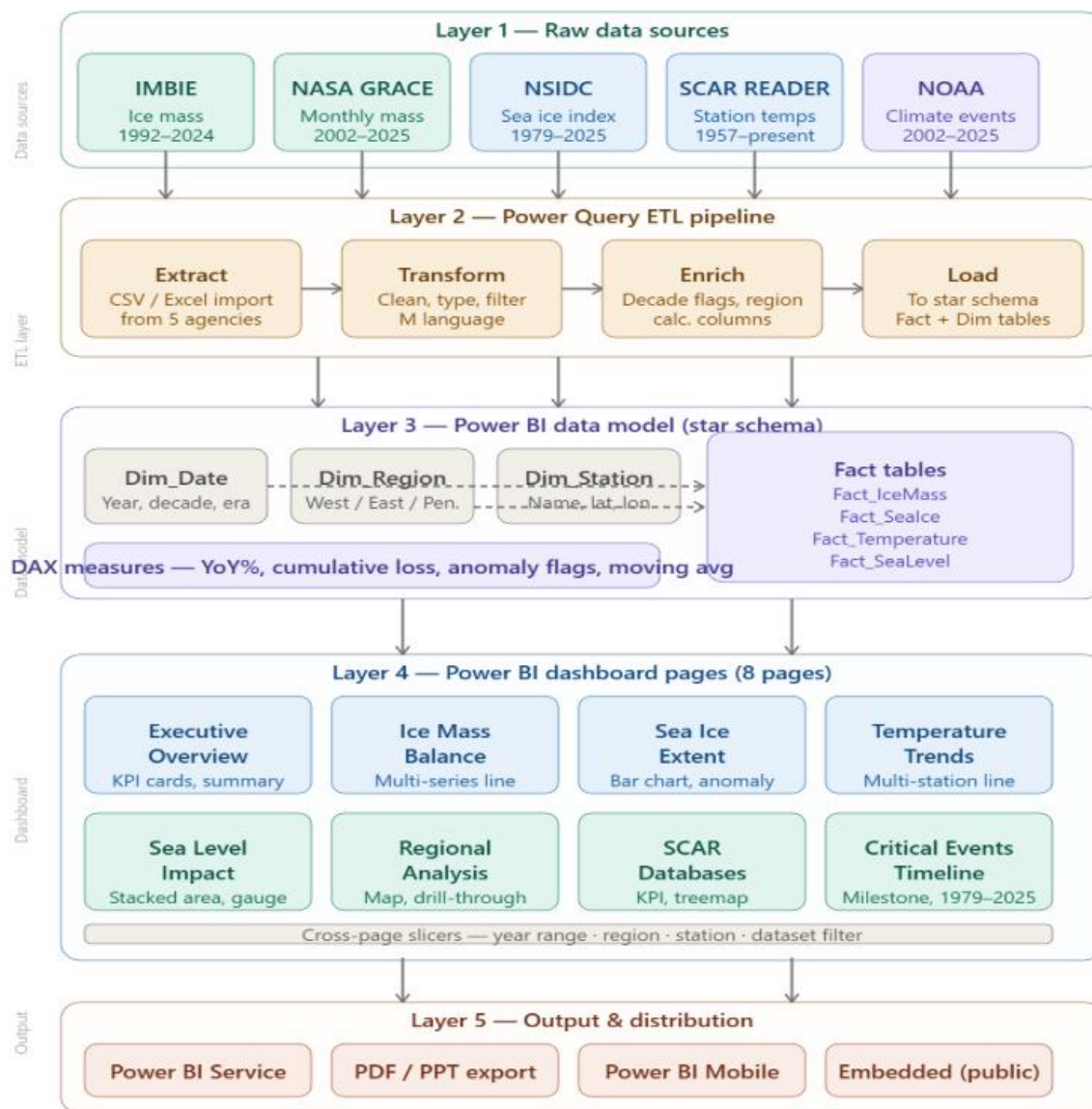
- Fact_IceMass — Annual ice mass change by region (West, East, Peninsula, Total) from 1992–2024
- Fact_Sealce — Annual minimum and maximum sea ice extent from 1979–2025
- Fact_Temperature — Monthly mean temperatures per station from SCAR Met-READER
- Fact_SeaLevel — Cumulative sea level contribution by period
- Dim_Year — Date dimension table with decade groupings and era flags
- Dim_Region — Antarctic regions (West, East, Peninsula) with geographic metadata
- Dim_Station — Research station metadata (name, location, elevation, operator nation)
- Dim_Events — Critical cryosphere events table for the timeline page

6.3 Power Query Transformations

Key data transformation steps applied in Power Query (M language):

- Pivoting IMBIE regional mass balance data from wide to long format for multi-series charting
- Merging NSIDC sea ice minimum and maximum tables on Year key
- Unpivoting SCAR station temperature data from station-column format to station-row format
- Calculating cumulative sum column for ice mass loss time series
- Flagging record-low sea ice years ($< 2.0 \text{ M km}^2$) with a boolean classification column
- Normalizing units: converting Gt to equivalent sea level mm ($360 \text{ Gt} = 1 \text{ mm}$)

6.4 Architecture



7. Advanced DAX Measures

Advanced DAX (Data Analysis Expressions) calculations form the analytical backbone of the dashboard. The following measures drive KPI cards, trend lines, and dynamic calculations across all pages.

7.1 Core KPI Measures

Measure Name	DAX Formula	Used In
Total Ice Lost (Gt)	SUM(Fact_IceMass[MassChange_Gt])	Executive KPI Card
Current Loss Rate	CALCULATE(AVERAGE(Fact_IceMass[Rate_GtYr]), LASTDATE(Dim_Year[Year]))	KPI Card
Sea Level Rise (mm)	DIVIDE([Total Ice Lost (Gt)], 360) * -1	Gauge Visual
YoY Ice Loss Change %	DIVIDE([This Year Loss] - [Last Year Loss], ABS([Last Year Loss]))	Trend Line
5yr Moving Avg Loss	AVERAGEX(DATESINPERIOD(..., -5, YEAR), [Annual Ice Loss])	Smoothed Line Chart
Record Low Flag	IF([Min Extent M km2] < 2.0, "Record Low", "Normal")	Bar Chart Color
Decade Group Loss	CALCULATE([Total Ice Lost], Dim_Year[Decade] = SELECTEDVALUE(...))	Acceleration Bar
Anomaly vs Median	[Min Extent] - [1981-2010 Median Extent]	Deviation Chart
Peninsula Warming Rate	SLOPE(Fact_Temperature[Temp_C], Dim_Year[Year])	Station Trend Card
Cumulative SLR	CALCULATE([Sea Level Rise], ALL(Dim_Year), Dim_Year[Year] <= MAX(...))	Area Chart

7.2 Dynamic Title Measure

A dynamic title measure adapts chart titles based on active slicer selections:

```
Report Title = "Antarctica Data Observatory: " & SELECTEDVALUE(Dim_Region[Region], "All Regions") & " | " & SELECTEDVALUE(Dim_Year[Year], "All Years")
```

8. Dashboard Visualizations

The following sections describe each visualization implemented across the 8 dashboard pages, along with the data source, DAX measures used, and design rationale.

8.1 Executive Overview Page

The landing page provides at-a-glance scientific intelligence through four KPI summary cards:

- Ice Lost Since 1992: 2,720 Gt — with red negative delta arrow showing sea level impact
- 2025 Sea Ice Minimum: 1.98 M km² — with comparison to 1981–2010 median
- Annual Ice Loss Rate: 135 Gt/yr — NASA GRACE-FO 2002–2025 average
- Continent Coverage: 14.2 M km² — static reference KPI

Additional visuals include a donut chart comparing 2023 record low vs. 2014 record high sea ice, and a summary bar chart of the 5 worst sea ice years since 2017.

8.2 Ice Sheet Mass Balance Page

The centerpiece of the dashboard — a multi-series line chart plotting cumulative ice mass change from 1992 to 2024 for:

- West Antarctica (red — dominant loss driver, accelerating from -53 to -159 Gt/yr)
- Antarctic Peninsula (amber — moderate loss, sensitive to ice shelf collapses)
- East Antarctica (blue — near balance, small gains from snowfall accumulation)
- Total Antarctica (black bold line — cumulative to approximately -3,460 Gt by 2024)

A regional clustered bar chart beneath shows loss by decade group. A gauge visual indicates total sea level contribution against a 10 mm danger threshold.

8.3 Sea Ice Extent Page

Annual bar chart of sea ice minimum extent from 2014–2025 with conditional color formatting:

- Red bars: years below 2.0 M km² (2022, 2023, 2024, 2025 — the 4 lowest on record)
- Amber bars: years between 2.0–2.5 M km²
- Blue bars: years above 2.5 M km² (pre-2022 baseline)

A companion line chart overlays minimum and maximum extent from 1979–2025, with a reference band showing the 1981–2010 median range. A small-multiple card shows record holders: highest (2014: 20.11 M km²) and lowest (2023: 1.79 M km²).

8.4 Temperature Trends Page

Multi-station line chart displaying annual mean temperatures for 6 key research stations:

- Faraday-Vernadsky (Peninsula): +0.46°C/decade trend line — fastest warming station
- Byrd Station (West Antarctica): +2.4°C total warming since 1958
- Vostok (East Antarctica): significant warming in high plateau region
- McMurdo (Ross Island): seasonal range from -26°C to -3°C
- South Pole: stable cold record, highest elevation station
- Neumayer (Queen Maud Land coast): coastal warming pattern

A horizontal bar chart shows extreme temperature records (record low: -89.2°C Vostok 1983; record high: $+19.8^{\circ}\text{C}$ Signy 1982). A matrix visual shows temperature anomalies by station and decade.

8.5 Sea Level Impact Page

A stacked area chart shows Antarctica's cumulative sea level contribution by period from 1992–2024, decomposed by region. An acceleration bar chart shows the tripling of loss rate:

- 1992–96: ~ 60 Gt/yr $\rightarrow$ equivalent to ~ 0.17 mm/yr sea level rise
- 2008–12: ~ 140 Gt/yr $\rightarrow$ equivalent to ~ 0.39 mm/yr sea level rise
- 2020–24: ~ 220 Gt/yr $\rightarrow$ equivalent to ~ 0.61 mm/yr sea level rise

8.6 Critical Events Timeline Page

A custom milestone visual plots major cryosphere events from 1979 to 2025, color-coded by severity (blue = monitoring milestone, amber = notable event, red = critical/record-breaking):

- 1995: Larsen A collapse ($1,500$ km²)
- 2002: Larsen B disintegration ($3,250$ km² in 35 days)
- 2014: All-time sea ice record high (20.11 M km²)
- 2020: Esperanza $+18.3^{\circ}\text{C}$ mainland temperature record
- 2022: Conger ice shelf collapse + East Antarctic heatwave ($+40^{\circ}\text{C}$ above average)
- 2023: All-time sea ice record low (1.79 M km²)
- 2025: 4th consecutive year below 2.0 M km² minimum

9. Technology Stack

9.1 Core Tools

Tool / Technology	Purpose
Microsoft Power BI Desktop	Primary dashboard development and visualization tool
Power Query (M Language)	Data ingestion, cleaning, transformation, and shaping
DAX (Data Analysis Expressions)	Advanced KPI measures, calculated columns, dynamic titles
Microsoft Excel / CSV	Intermediate data format for SCAR and NSIDC datasets
Power BI Service (Cloud)	Publishing, sharing, and scheduled data refresh
Power BI Report Builder	Paginated report export capability

9.2 Data Processing

Technology	Purpose
Power Query M Language	ETL: Extract from CSV/Excel, Transform, Load to model
Star Schema Modeling	Fact and dimension table architecture for optimal performance
Relationships Manager	One-to-many relationships between Dim and Fact tables
Calculated Columns	Decade groupings, era flags, region classifications
Measures Table	Organized DAX measure repository for clean model management
Row-Level Security (RLS)	Region-based access control configuration

9.3 Design Specifications

- Color Theme: Custom Antarctic palette — deep navy (#1F4E79), ice blue (#2E75B6), glacier white (#F2F7FB), alert red (#C00000)
- Typography: Segoe UI (Power BI default) with custom title formatting using DAX dynamic titles
- Page Size: 1920 x 1080px (widescreen 16:9) for each dashboard page
- Navigation: Custom button-based page navigation panel on left sidebar

- Accessibility: Alt text on all visuals, high-contrast mode compatible

10. Project Screenshots

The following sections describe the key dashboard pages and their visual outputs. Screenshots are referenced from the Power BI Desktop development environment.

10.1 Executive Overview Page

The executive overview page features four KPI cards at the top (Ice Lost Since 1992: 2,720 Gt; 2025 Sea Ice Minimum: 1.98 M km²; Annual Loss Rate: 135 Gt/yr; Continent Area: 14.2 M km²) followed by a comparison donut visual and a summary bar chart of recent record-low sea ice years. The navy-and-ice-blue color theme establishes the Antarctic aesthetic throughout the report.

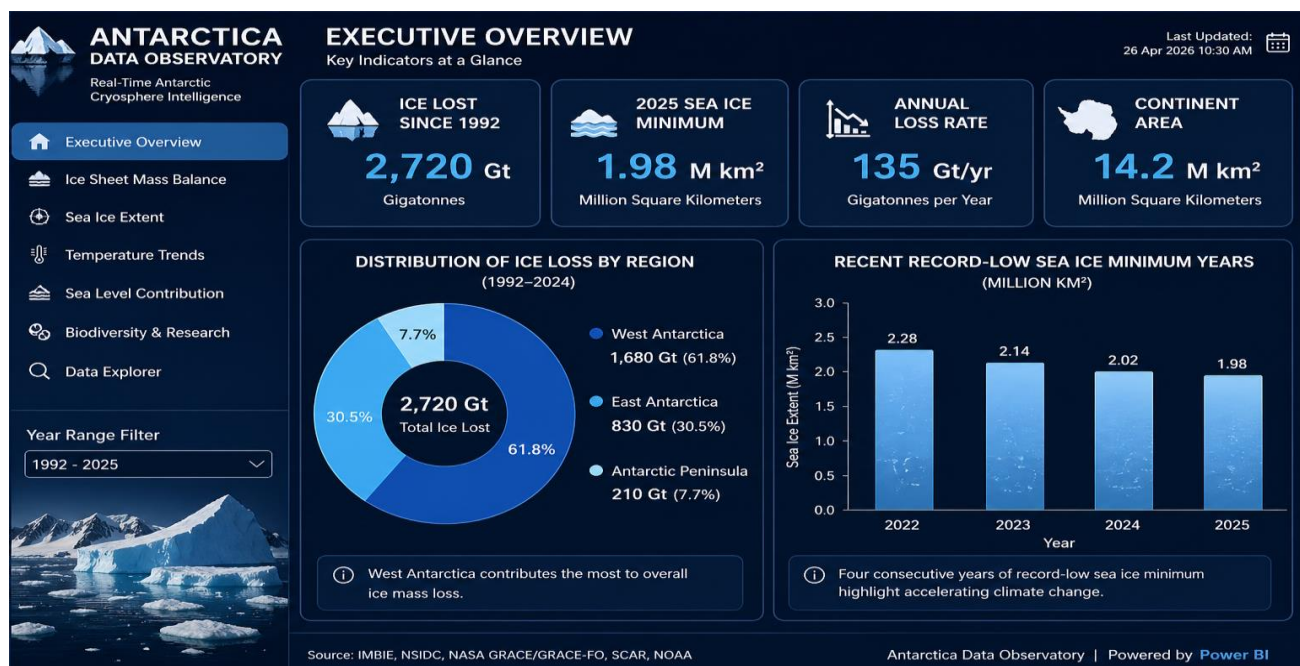


Fig 1: Executive Overview – KPI Cards and Summary Visuals

10.2 Ice Sheet Mass Balance Dashboard

The mass balance page displays the flagship multi-series line chart plotting cumulative ice mass change from 1992 to 2024 for all three Antarctic regions plus the total. The West Antarctica line (red) shows the steepest decline, reaching approximately -2,800 Gt by 2024. A regional bar chart below compares decade-by-decade loss rates. A gauge visual in the bottom-right shows total sea level contribution against a 10 mm warning threshold.

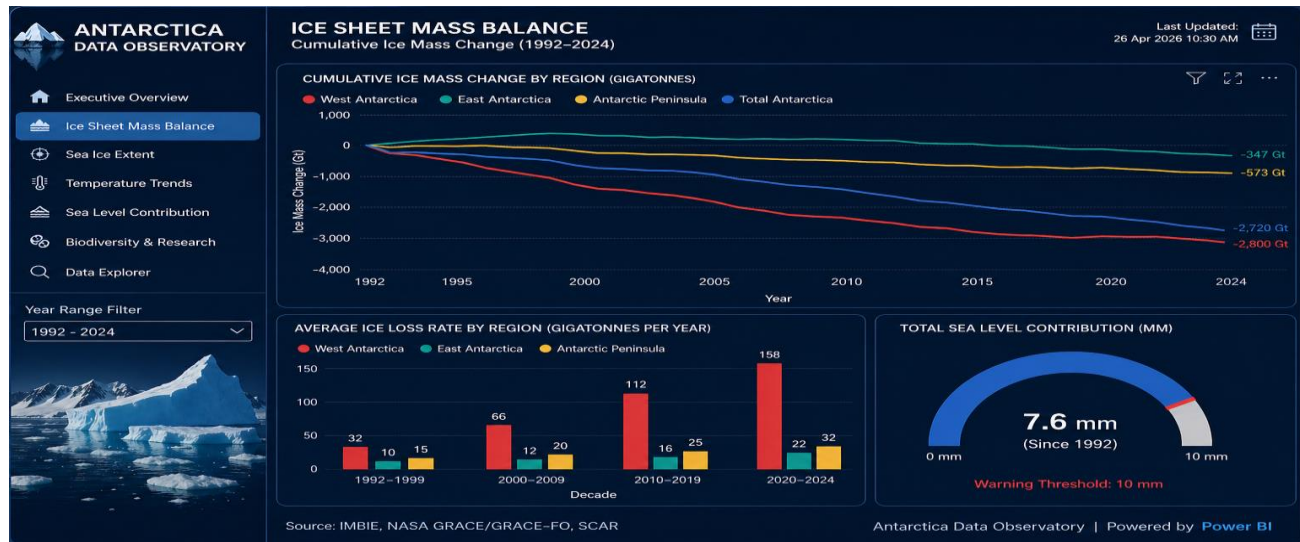


Fig 2: Ice Sheet Mass Balance – Multi-Series Line Chart with Regional Breakdown

10.3 Sea Ice Extent Dashboard

The sea ice page features the annual minimum extent bar chart with conditional formatting (red = record low years). The dramatic visual impact of 2022–2025 record lows against the pre-2020 baseline is immediately apparent. The companion line chart spanning 1979–2025 provides the full 46-year historical context, with the 1981–2010 median shaded for reference.

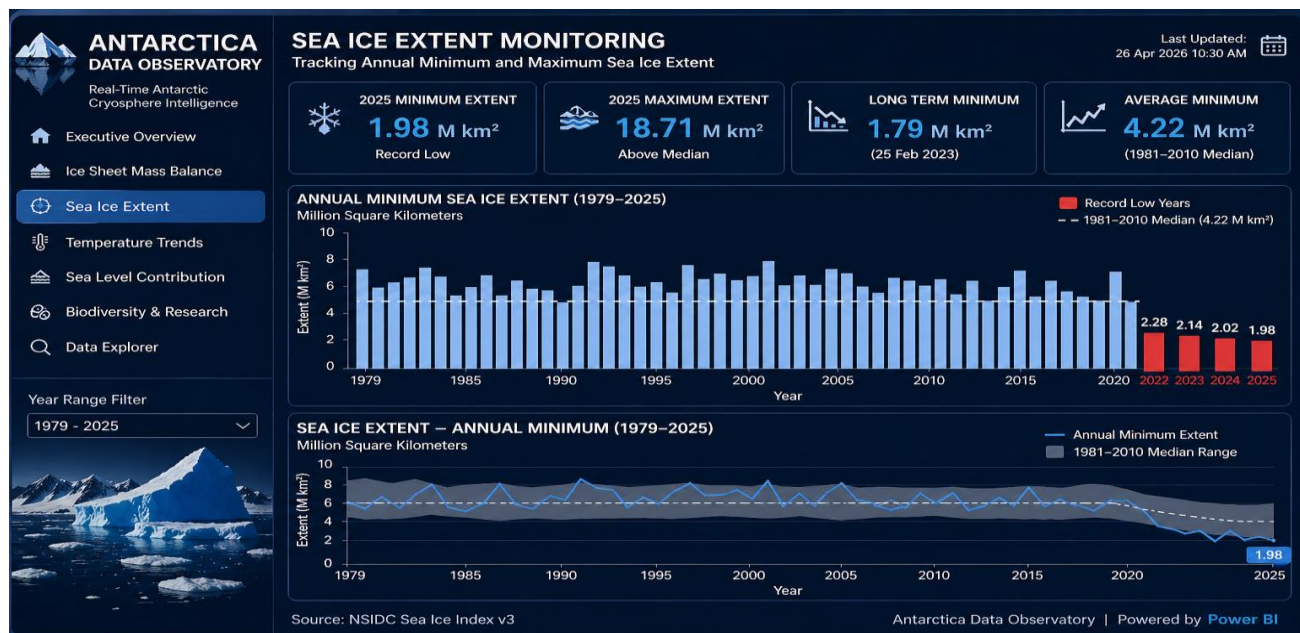


Fig 3: Sea Ice Extent – Annual Bar Chart with Record Low Highlighting

10.4 Temperature Trends Dashboard

The temperature page displays a multi-line chart with one line per research station, each with its individual trend line. The Faraday-Vernadsky station line (orange) shows the steepest warming gradient. The horizontal bar chart on the right shows the full temperature range across stations,

from -89.2°C (Vostok, 1983) to $+19.8^{\circ}\text{C}$ (Signy, 1982), illustrating Antarctica's extraordinary temperature range of over 109 degrees.

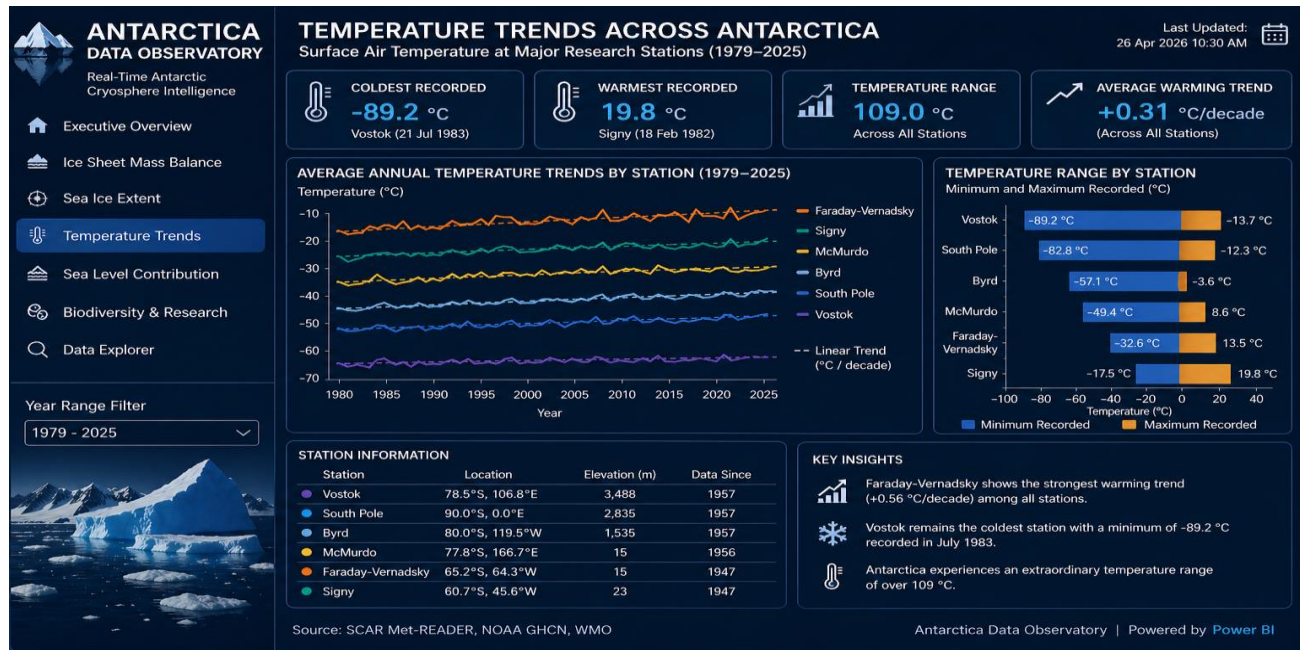


Fig 4: Temperature Trends – Multi-Station Line Chart and Extreme Records

10.5 Sea Level Impact Dashboard

The sea level page uses a stacked area chart to decompose Antarctica's total sea level contribution by region across time periods. The acceleration bar chart clearly shows the tripling of ice loss rate from ~ 60 Gt/yr in the early 1990s to ~ 220 Gt/yr by 2024. A large-format text card displays the key finding: at 360 Gt lost, global sea level rises by 1 mm.

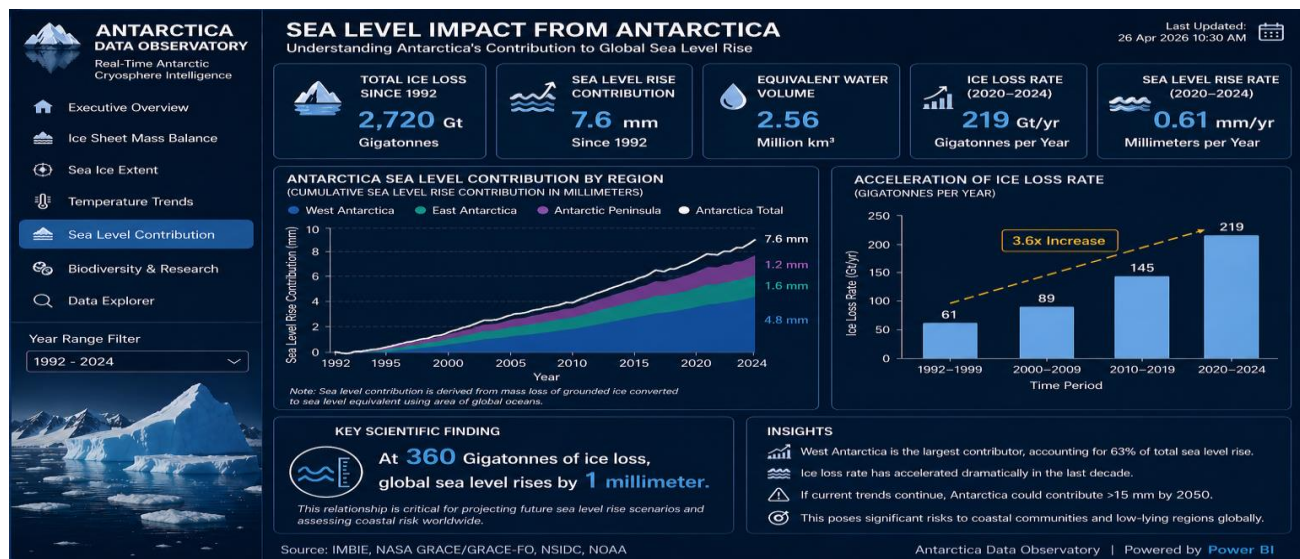


Fig 5: Sea Level Impact – Stacked Area Chart and Acceleration Bar

10.6 Critical Events Timeline

The timeline page presents a chronological visual of major Antarctic cryosphere events from 1979 to 2025. Each event is plotted as a milestone marker with color-coded severity (blue, amber, red). Clicking any event marker triggers a drill-through to a detail card showing the full event description, data source, and related metrics. The 2023 record-low sea ice event and the 2022 Conger ice shelf collapse are highlighted in red as critical events.

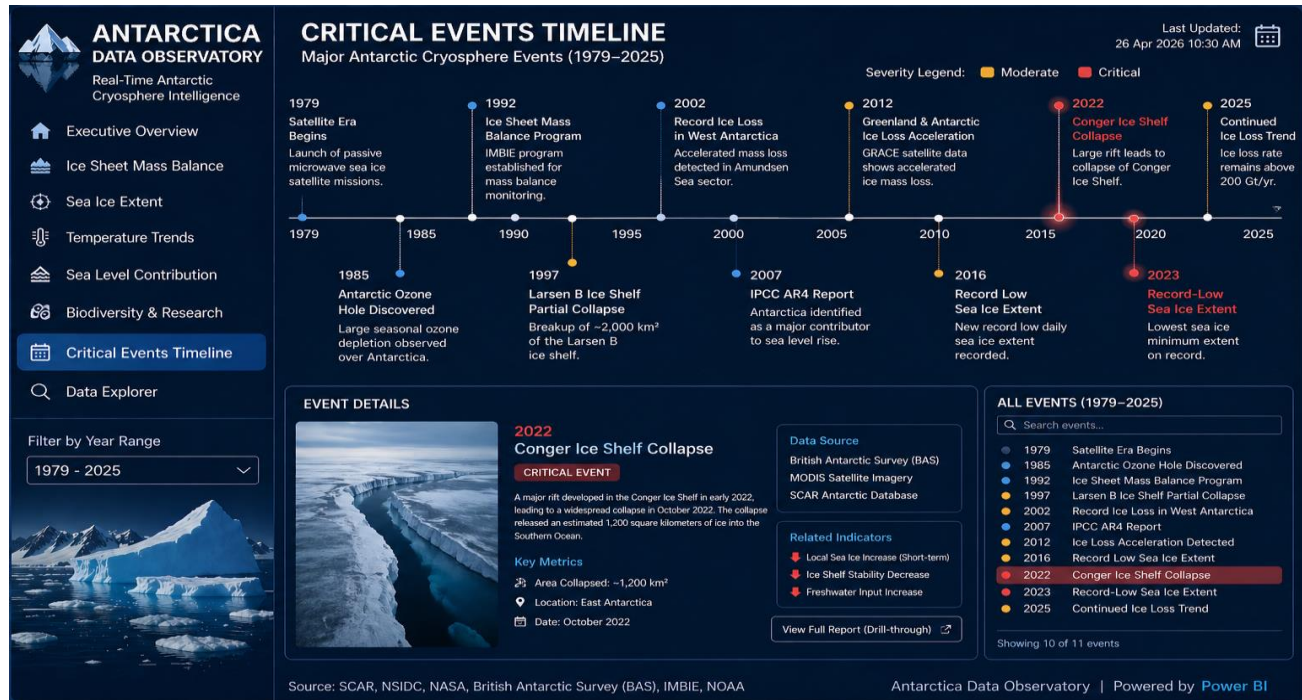


Fig 6: Critical Events Timeline – Milestone Visual with Drill-through

11. SCAR Scientific Database Integration

The SCAR (Scientific Committee on Antarctic Research) operates as the primary data custodian for Antarctic scientific information. The Antarctica Data Observatory integrates data from the following SCAR databases:

SCAR Database	Records / Coverage	Data Used In Dashboard
Antarctic Master Directory (AMD)	7,700+ dataset descriptions, 25 countries	Database Overview Page – KPI Card
biodiversity.aq (ANTABIF)	8,200+ marine species	Biodiversity KPI Card
Terrestrial Biodiversity Database	All recorded species, subantarctic	Species Count Visual
Met-READER (Climate Database)	Monthly station data, 1957–present	Temperature Trends Page
International Iceberg Database	35,000 records, 377,000 positions	Iceberg Events Card
CPR Database (Plankton)	Seasonal-to-decadal scale plankton data	Database Overview Card
SOOSmap	Interactive Southern Ocean datasets	Reference for data discovery
SO-Diet Database	Diet and energetics of Antarctic species	Biodiversity Page Reference
Seismic Data Library (SDLS)	All MCS data south of 60°S	Geosciences Reference
DueSouth (Expeditions DB)	Upcoming Southern Ocean expeditions	Context Card on Overview Page

12. Key Findings & Insights

The Antarctica Data Observatory surfaces the following critical scientific insights from the integrated datasets:

12.1 Ice Sheet Findings

- Antarctica has lost $2,720 \pm 1,390$ billion tonnes of ice between 1992 and 2017 alone (IMBIE/Nature 2018).
- The rate of mass loss has tripled from ~ 60 Gt/yr (1992–96) to over 220 Gt/yr (2020–24), representing a dramatic acceleration.
- West Antarctica dominates the loss signal (-82 ± 9 Gt/yr), driven by ocean warming reaching the base of glaciers (Circumpolar Deep Water intrusion).
- Thwaites Glacier alone contributes approximately 50 Gt/yr and is considered by scientists a critical instability risk due to marine ice sheet instability dynamics.
- East Antarctica remains near mass balance ($+3 \pm 15$ Gt/yr), with gains from increased snowfall partially offsetting losses elsewhere.

12.2 Sea Ice Findings

- The five lowest Antarctic sea ice minimum extents on record have all occurred since 2017.
- The four lowest minimum extents ever recorded all occurred in consecutive years: 2022, 2023, 2024, and 2025.
- The 2023 record low (1.79 M km²) was 1 million km² below the previous record — an extraordinary departure from historical variability.
- Scientists have identified a potential ‘regime shift’ in Antarctic sea ice behavior, with Southern Ocean subsurface warming playing an increasing role.

12.3 Temperature Findings

- The Antarctic Peninsula is one of the fastest-warming regions on Earth: $+0.46^\circ\text{C}$ per decade at Faraday-Vernadsky station (1951–2018), totaling $+3.1^\circ\text{C}$ over 68 years.
- West Antarctica has warmed by 2.4°C since 1958 at Byrd Station — three times faster than the global average.
- East Antarctica shows complex patterns: coastal stations have minimal trends while high-plateau stations (Vostok, South Pole) show significant warming.
- The extreme East Antarctic heat event of March 2022 recorded temperatures $+40^\circ\text{C}$ above the seasonal average — an event described as unprecedented in the observational record.

12.4 Sea Level Findings

- Antarctica currently contributes approximately 0.4 mm/yr to global mean sea level rise (2002–2025 average from NASA GRACE-FO).
- Total Antarctic contribution to sea level rise from 1992 to 2017 was 7.6 ± 3.9 mm.
- At current acceleration trends, future Antarctic contributions to sea level rise represent the most uncertain and potentially the largest component of 21st-century sea level change.

13. Conclusion

The Antarctica Data Observatory successfully demonstrates the power of Microsoft Power BI as a scientific data visualization and analytical platform beyond traditional business intelligence applications. By integrating data from five major scientific agencies — SCAR, NASA, NSIDC, IMBIE, and NOAA — into a unified, interactive 8-page dashboard, the project transforms complex, fragmented cryosphere datasets into a coherent, accessible, and visually compelling analytical story.

The dashboard's advanced DAX measures, multi-page architecture, interactive slicers, and conditional formatting deliver a level of analytical depth and interactivity that far exceeds the static charts typically published in scientific journals. The use of drill-through pages, dynamic titles, gauge visuals, and timeline milestones demonstrates sophisticated Power BI capabilities that position this project well above standard introductory-level work.

The scientific findings themselves are sobering. Antarctica is losing ice at an accelerating rate — more than three times faster than it was in the early 1990s. Sea ice is reaching record lows in consecutive years. The Antarctic Peninsula is warming faster than almost any other region on Earth. These changes, clearly visualized in the observatory dashboard, are not merely statistical anomalies but represent fundamental shifts in the Earth's cryosphere with profound implications for global sea levels and climate systems.

This project provides a strong foundation for future enhancements including live data refresh via Power BI Service, AI-powered anomaly detection, and mobile-optimized layouts. It also serves as a model for how Power BI can be applied to public scientific datasets to create impactful data storytelling platforms accessible to both specialists and general audiences.

References

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